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1. Introduction

1.1 Project Overview and Objectives :

Overview

Elite Motors is a comprehensive web-based car dealership platform that connects car buyers and sellers. The system provides an intuitive interface for users to browse, search, and purchase vehicles, while also offering a platform for individuals to sell their cars.

Objectives

1. User-Friendly Interface

- Create an intuitive and responsive design for seamless navigation
- Implement easy-to-use search and filter functionality
- Ensure mobile responsiveness for all device types

2. Vehicle Management

- Display detailed car listings with images and specifications
- Categorize vehicles by type (Sedan, SUV, etc.)
- Provide comprehensive vehicle information including price, mileage, and features

3. User Engagement

- Enable users to browse available inventory
- Allow users to contact the dealership
- Provide a platform for selling used cars

4. Admin Control

- Manage vehicle inventory
- Handle user inquiries and messages
- Update car listings and availability

5. Security

- Secure user data and transactions
- Implement admin authentication
- Protect against common web vulnerabilities

6. Performance

- Ensure fast page load times
- Optimize image loading
- Implement efficient database queries

System Design

1.2 Technology Stack Used :

2.1 Technology Stack

Component	Technologies Used
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5
Backend	PHP 7.4+
Database	MySQL 5.7+
Server	XAMPP (Apache, MySQL, PHP)
APIs	RESTful API endpoints
Version Control	Git

2.2 System Architecture

Layer	Components
Presentation	Responsive Web Interface (HTML/CSS/JS)
Application	PHP scripts handling business logic
Data Access	MySQL Database
Server	Apache Web Server (XAMPP)

2.3 Database Schema

Table Name	Description	Key Fields
Users	Stores user account information	user_id, username, email, password
cars	Contains vehicle listings	car_id, make, model, year, price, etc.
messages	Stores contact form submissions	msg_id, name, email, subject, message
car_images	Stores images related to car listings	image_id, car_id, image_path

3. Project Modules

3.1 User Module

Feature	Description
Car Browsing	View available cars with filters by type, price range, and specifications
Car Purchase	View car details and contact for purchase
Sell Car	Form for users to submit their car for sale

3.2 Admin Module

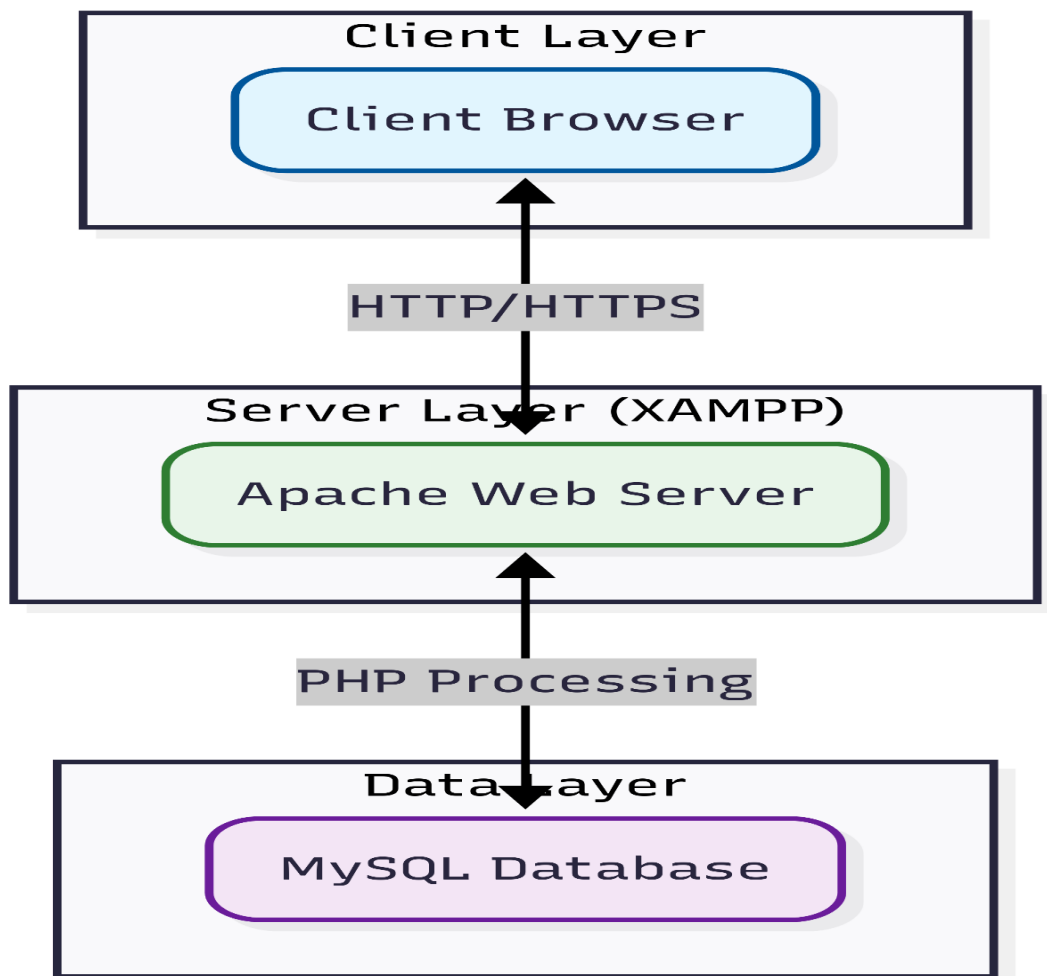
Feature	Description
Car Management	Add, edit, or remove car listings
User Management	Manage user accounts and permissions

Feature	Description
Message Handling	View and respond to customer inquiries

3.3 Database Module

Component	Description
Connection	Handles database connections and queries
Queries	Predefined SQL queries for data manipulation
Data Validation	Ensures data integrity and security

2.2 System Architecture



❖ Key Components:

1. Client Layer

- Web Browsers (Chrome, Firefox, etc.)
- Responsive UI Components

2. Server Layer

- Apache Web Server
- PHP Scripts
- Request Handler

3. Data Layer

- MySQL Database
- Data Storage
- File System (for images)

1.3 Database Schema :

ables Overview

1. users

- user_id (PK)
- username
- email
- password (hashed)
- user_type (admin/customer)

2. cars

- car_id (PK)
- make
- model
- year
- price
- mileage
- transmission
- fuel_type
- status (available/sold)

3. car_images

- image_id (PK)
- car_id (FK)
- image_path
- is_primary (boolean)

4. messages

- message_id (PK)
- name
- email

- subject
- message
- created_at

Relationships:

- One car can have multiple images (1:N)
- Users can send multiple messages (1:N)

3. Project Modules

3.1 User Module

Feature	Components	Description
Car Browsing		
	Category View	Browse by vehicle type (Sedan, SUV, Sports)
	Search & Filter	Filter by price, make, model, year, mileage
	Sorting	Sort by price, newest, mileage
Car Purchase		
	Vehicle Details	Specifications, features, history
	Inquiry System	Contact forms, test drive scheduling
	Favorites	Save preferred vehicles
Sell Car		
	Listing Creation	Form for vehicle details and images
	Price Calculator	Suggested pricing based on market data
	Status Tracking	View listing performance

3.2 Admin Module

Feature	Components	Description
Car Management		
	Inventory Control	Add/Edit/Remove vehicles
	Bulk Operations	Import/Export listings
	Promotions	Create special offers
User Management		
	Accounts	Create/Manage user profiles
	Permissions	Role-based access control
	Activity Logs	Track user actions
Message Handling		

Feature	Components	Description
	Inbox	View customer inquiries
	Templates	Predefined responses
	Notifications	Alerts for new messages

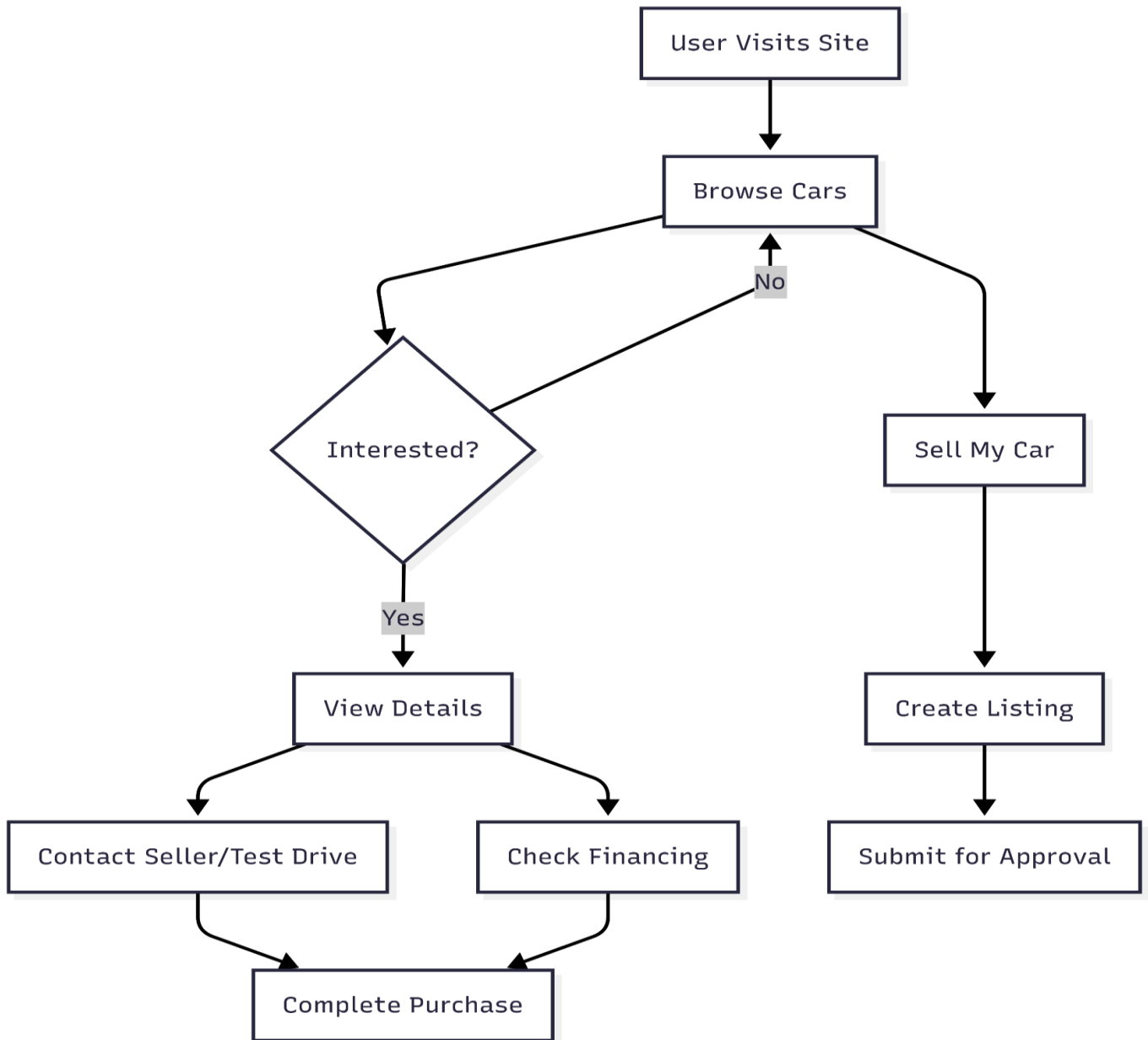
3.3 Database Module

Component	Functions	Description
Core Operations		
	CRUD	Create, Read, Update, Delete operations
	Transactions	Data consistency management
	Backups	Automated data backups
Security		
	Validation	Input sanitization
	Encryption	Secure data storage
	Access Control	Permission management
Performance		
	Indexing	Query optimization
	Caching	Frequently accessed data
	Optimization	Query performance tuning

4. System flow diagram

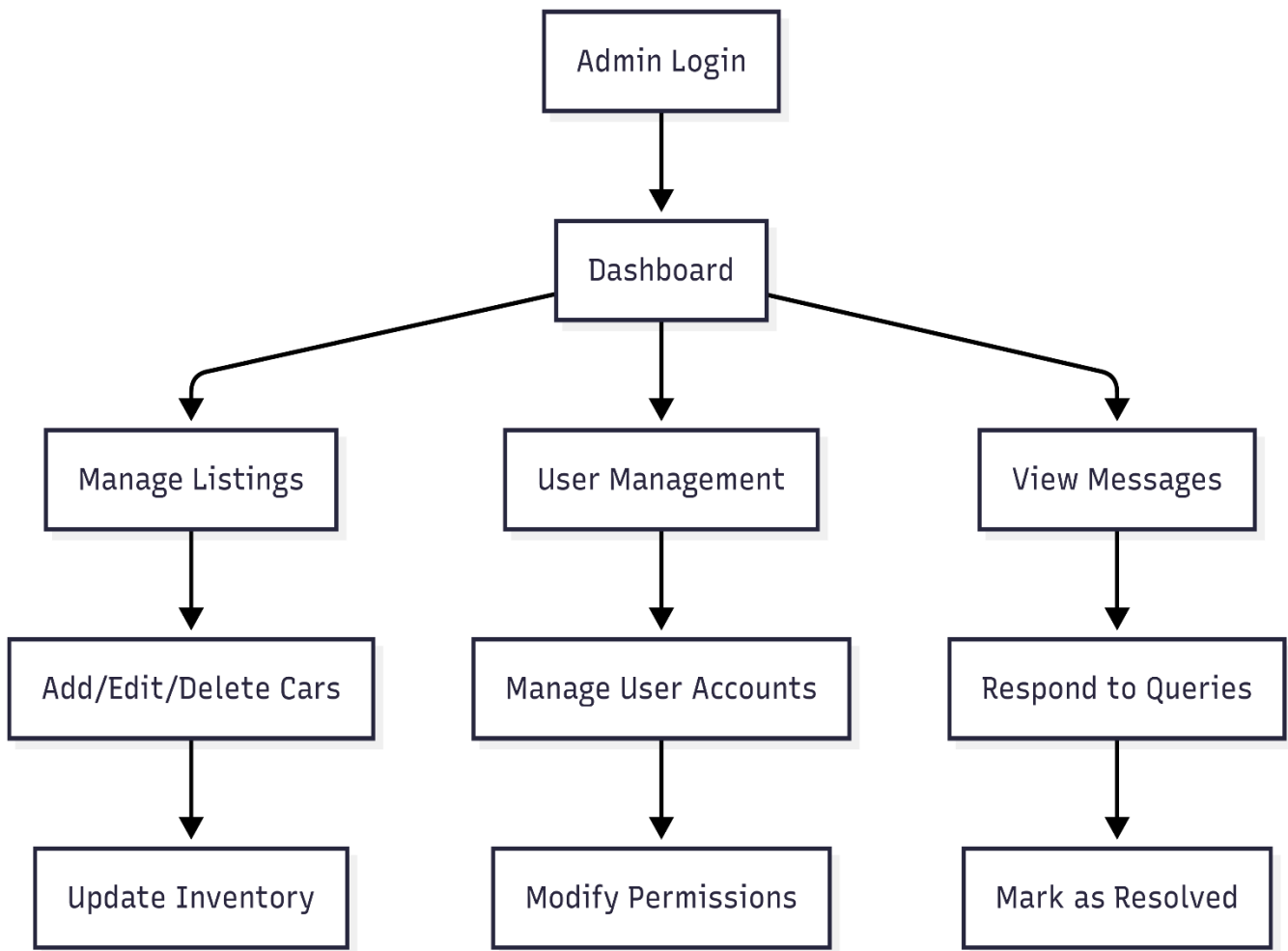
4.1 User Flow

- **User Flow:** Shows the customer journey from landing to purchase



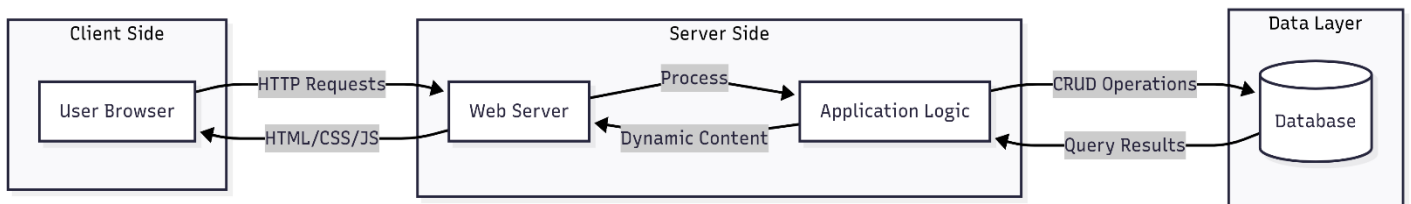
4.2 Admin Flow

- **Admin Flow:** Illustrates administrative functions and controls



4.3 Data Flow Diagram

- **Data Flow:** Demonstrates how data moves through the system



5. Implementation

5.1 Key Features

Car Listing

- **Dynamic Grid View**
 - Responsive card-based layout
 - Lazy loading for images
 - Category-based filtering
 - Sort by price, year, or mileage

Search and Filter

- **Advanced Search**
 - Make/Model search
 - Price range slider
 - Year selection
 - Transmission type
 - Fuel type
 - Mileage range

Contact Forms

- **Inquiry System**
 - Car-specific inquiry forms
 - Test drive scheduling
 - Trade-in valuation requests
 - Newsletter subscription

Admin Dashboard

- **Management Console**
 - Real-time inventory overview
 - Sales analytics
 - User management

- Message center
- Content management

5.2 File Structure

Root Directory

File/Directory	Purpose
index.php	Homepage with featured cars
buy_car.php	Car listing and details
sell_car.php	Sell car form
contact.php	Contact form handler
assets/	Static files
admin/	Admin panel files
database/	Database scripts
includes/	Reusable components

Admin Directory

File	Purpose
dashboard.php	Admin dashboard
cars/	Car management
users/	User management
messages/	Message center
reports/	Sales reports

Database Directory

File	Purpose
config.php	Database connection
schema.sql	Database schema
seeds/	Sample data
migrations/	Database migrations

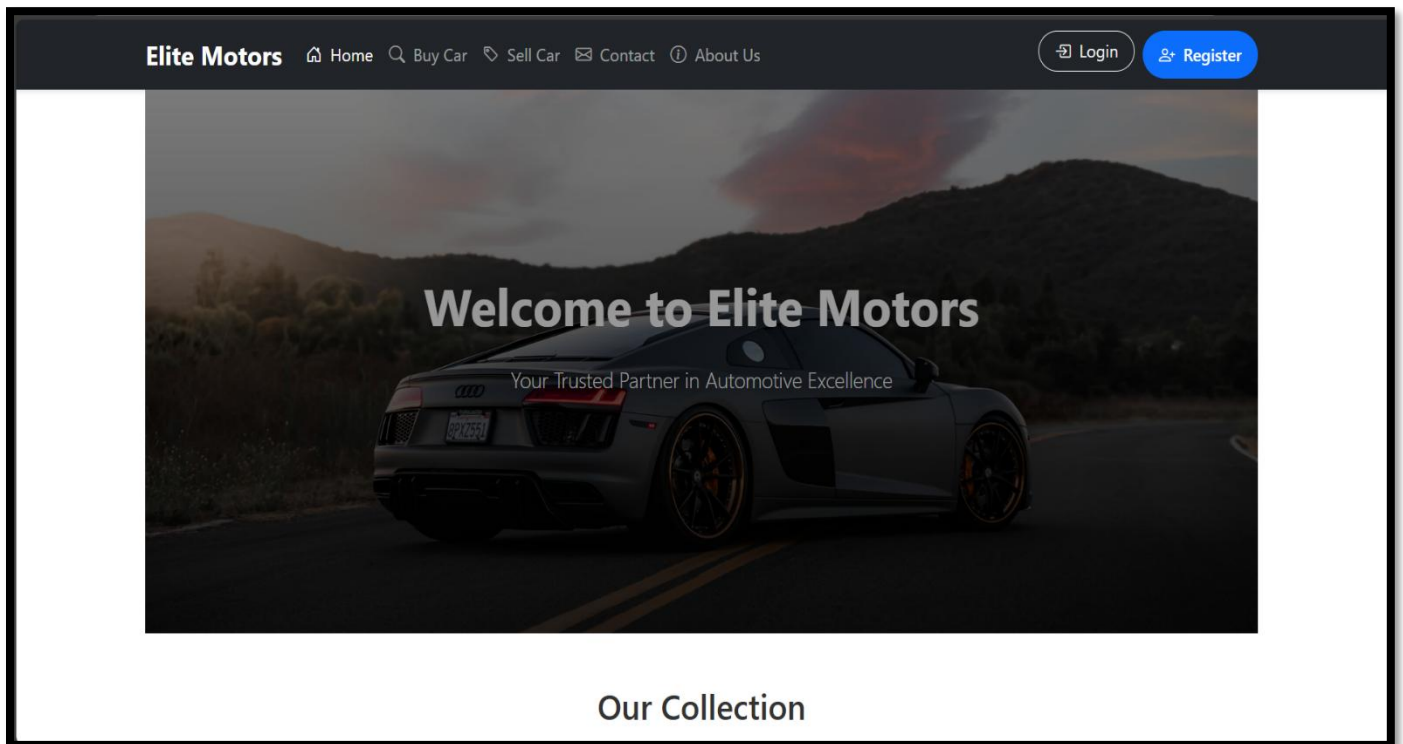
Assets Directory

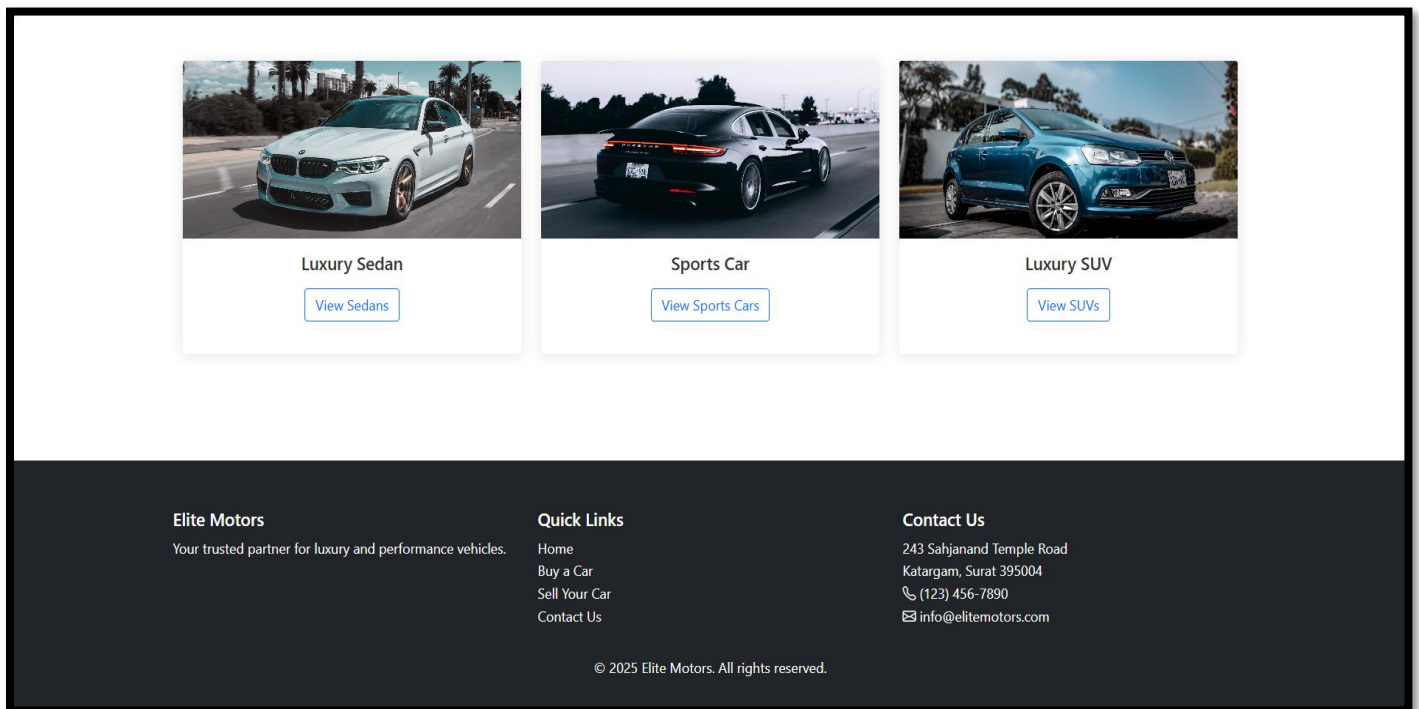
Directory	Contents
css/	Stylesheets
js/	JavaScript files
images/	Static images
uploads/	User-uploaded content

6. Screenshots

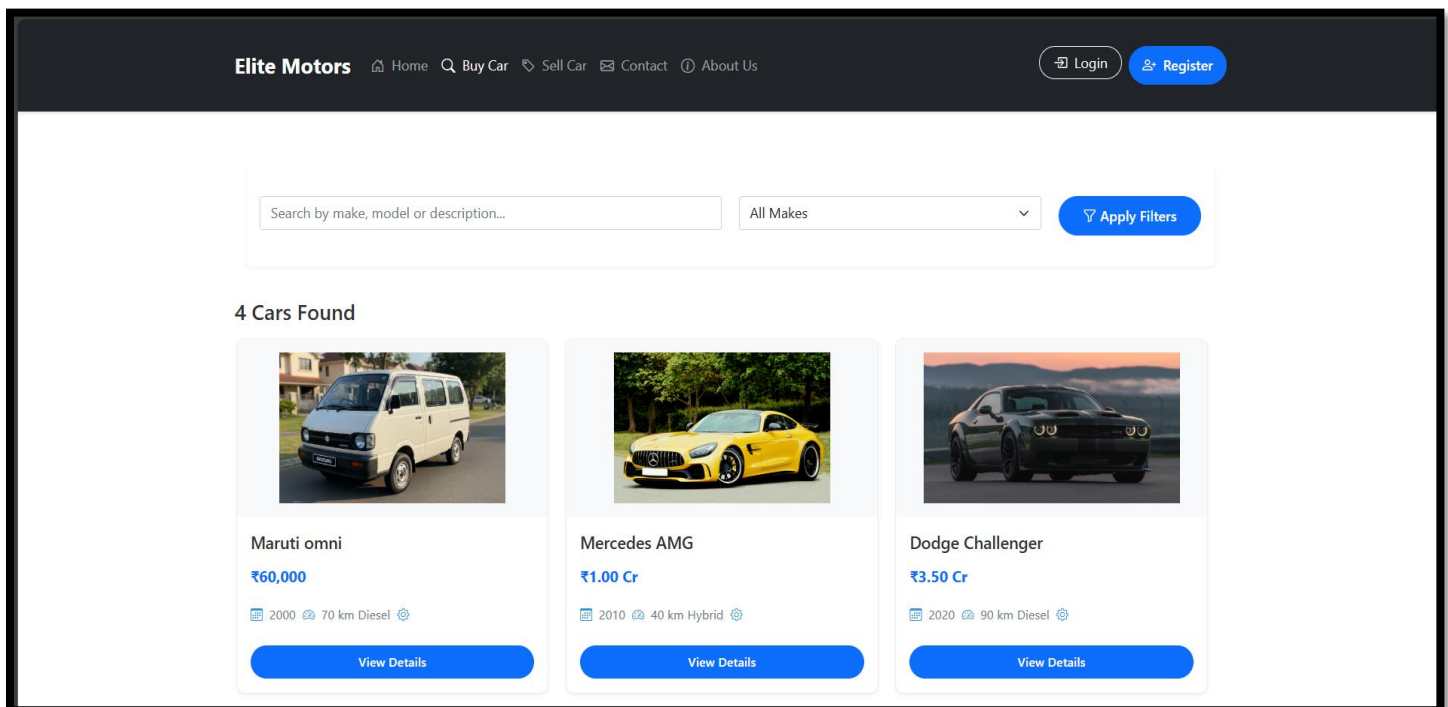
6.1 Home Page

- Client Website Screenshots:





■ Buy car page





Lamborghini Huracan H2

₹10.00 Cr

2022 50 km Petrol Automatic

[View Details](#)

Elite Motors

Your trusted partner for luxury and performance vehicles.

Quick Links

[Home](#)
[Buy a Car](#)
[Sell Your Car](#)
[Contact Us](#)

Contact Us

243 Sahjanand Temple Road
Katargam, Surat 395004
(123) 456-7890
info@elitemotors.com

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■ Car Details Page

[Home](#) / [Cars for Sale](#) / 2020 Dodge Challenger

₹35,000,000

[Contact Seller](#)[Overview](#)[Specs](#)

Vehicle Overview

3.6L Pentastar V6: The entry-point to the Charger lineup, providing a balance of performance and efficiency.

5.7L HEMI V8: A popular choice, delivering the classic V8 rumble and robust acceleration.

Vehicle History

Year: 2020

Mileage: 90 miles

Fuel Type: Diesel

Transmission:

Color: 0

Views: 0

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Quick Links

[Home](#)[Buy a Car](#)[Sell Your Car](#)[Contact Us](#)

Contact Us

243 Sahjanand Temple Road

Katargam, Surat 395004

(123) 456-7890

info@elitemotors.com

▪ Contact Seller Page/form

The screenshot shows a web page for 'Elite Motors' with a dark theme. The top navigation bar includes 'Home', 'Buy Car', and 'Sell Car' links, along with 'Login' and 'Register' buttons. The main content area displays a car listing for a '2020 Dodge Challenger' with a price of ₹35,000,000. A modal form titled 'Contact Seller - 2020 Dodge Challenger' is open, featuring input fields for 'Your Name', 'Email Address', and 'Phone Number', and a text area for 'Your Message'. The message text reads: 'Hello, I'm interested in the 2020 Dodge Challenger (ID: 17). Please provide more details.' The form has a 'Close' button and a blue 'Send Message' button. The background shows a car image and a 'Contact Seller' button on the right.

Elite Motors Home Buy Car Sell Car Login Register

Home / Cars for Sale / 2020 Dodge Challenger

₹35,000,000

Contact Seller

Contact Seller - 2020 Dodge Challenger

Your Name

Email Address

Phone Number

Your Message

Hello, I'm interested in the 2020 Dodge Challenger (ID: 17). Please provide more details.

Close Send Message

Overview Specs

▪ User Register Page

Elite Motors

[Home](#) [Buy Car](#) [Sell Car](#) [Contact](#) [About Us](#)

[Login](#) [Register](#)

Create an Account

First Name

Last Name

Username

@

Letters, numbers, and underscores only.

Email

Phone Number

Password

Confirm Password

Register

Already have an account? [Login here](#)

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Quick Links

[Home](#)
[Buy a Car](#)
[Sell Your Car](#)
[Contact Us](#)

Contact Us

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Katargam, Surat 395004
[\(123\) 456-7890](#)
[info@elitemotors.com](#)

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▪ User Login page

Elite Motors

[Home](#) [Buy Car](#) [Sell Car](#) [Contact](#) [About Us](#)

[Login](#) [Register](#)

Login

Username *

Password *

Login

Don't have an account? [Register here](#)

Elite Motors

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Quick Links

[Home](#)
[Buy a Car](#)
[Sell Your Car](#)
[Contact Us](#)

Contact Us

243 Sahjanand Temple Road
Katargam, Surat 395004
[📞 \(123\) 456-7890](#)
[✉ info@elitemotors.com](#)

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▪ Sell Car Page

Sell Your Car

Car Details

Make *

Model *

Year *

Mileage (km)

Engine Size (L)

Transmission *

Fuel Type *

Color *

Price (₹) *

Description

Phone Number *

Email *

Car Photo *

Max file size: 5MB. Allowed formats: JPG, PNG, GIF, WebP

[List My Car](#)

▪ User Contact Us form

Contact Us

Your Name *

Email Address *

Phone Number (Optional)

Subject (Optional)

Your Message *

 Send Message

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Quick Links

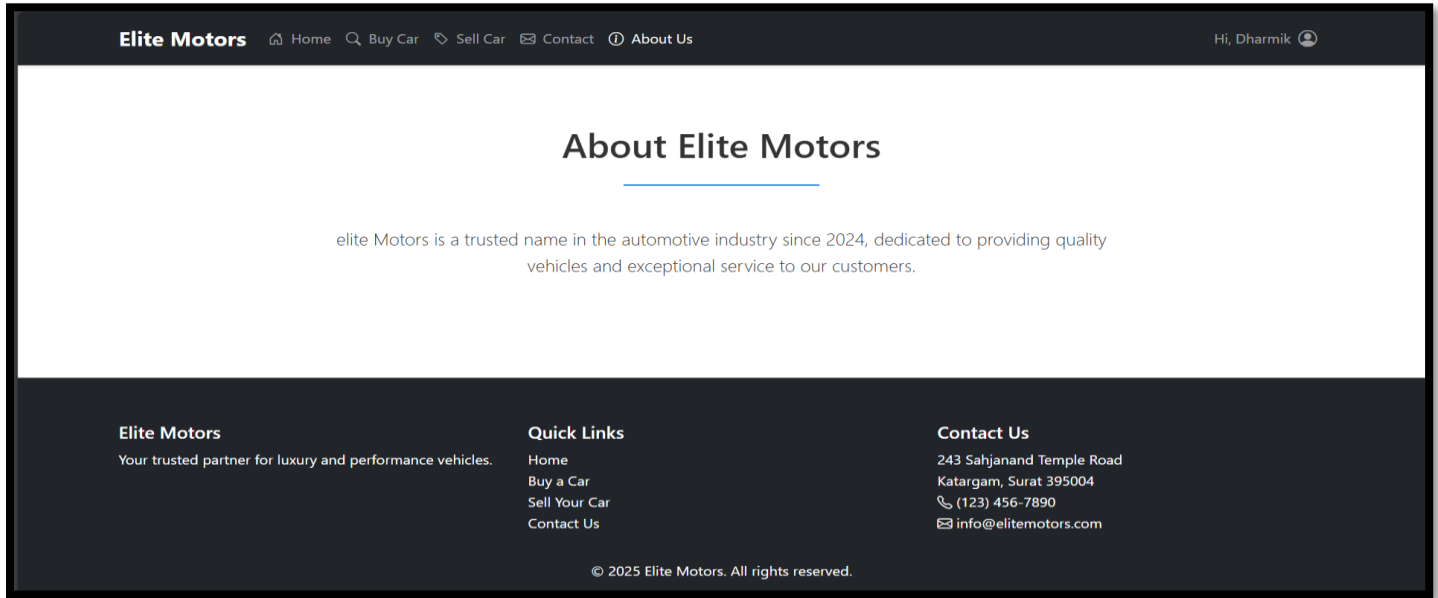
[Home](#)
[Buy a Car](#)
[Sell Your Car](#)
[Contact Us](#)

Contact Us

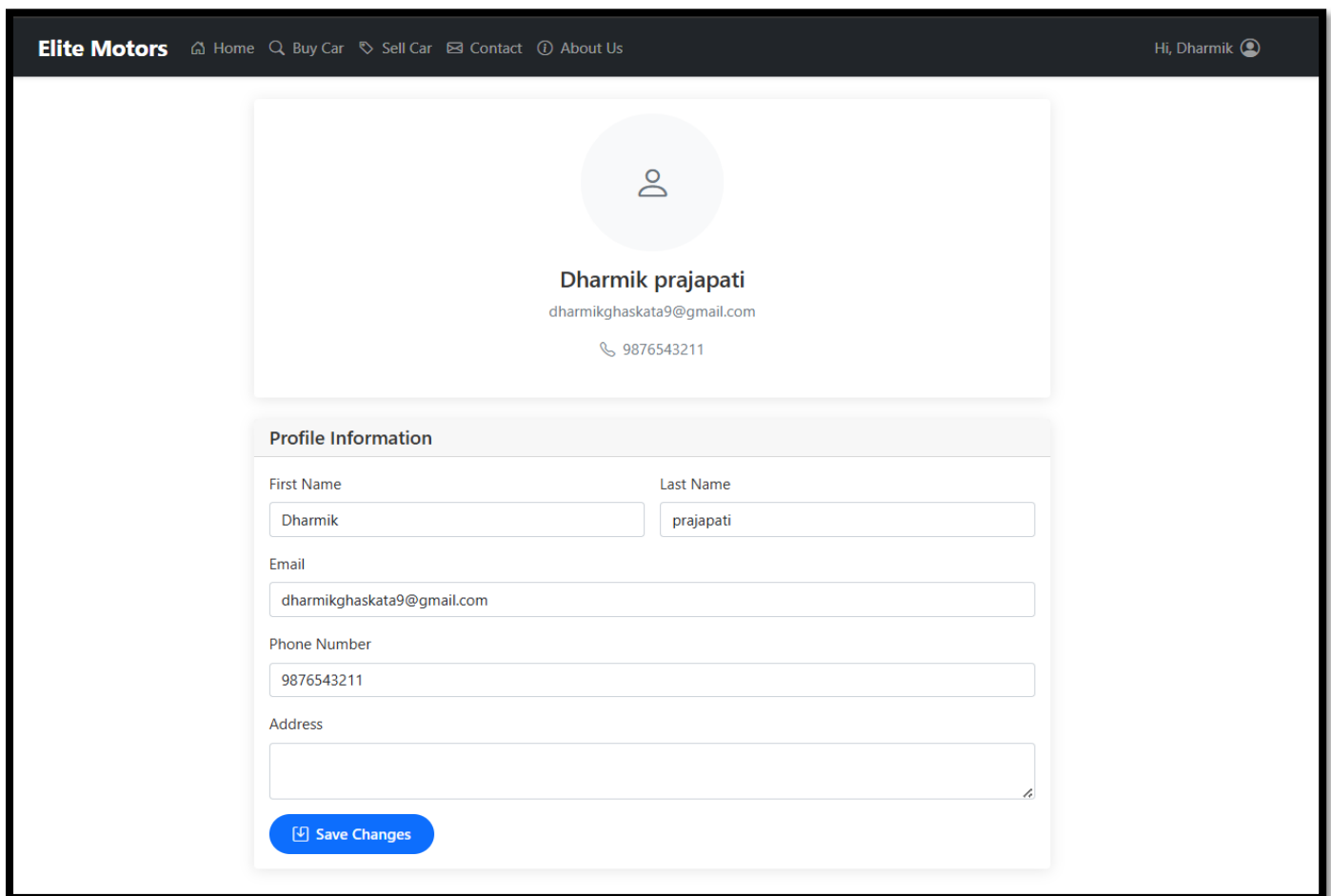
243 Sahjanand Temple Road
Katargam, Surat 395004
 (123) 456-7890
 info@elitemotors.com

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■ About Us page



■ User Profile Page



■ User Listing Page

Elite Motors

[Home](#) [Buy Car](#) [Sell Car](#) [Contact](#) [About Us](#)

Hi, yash

My Listings

List a New Car

TOTAL LISTINGS
5

ACTIVE
5

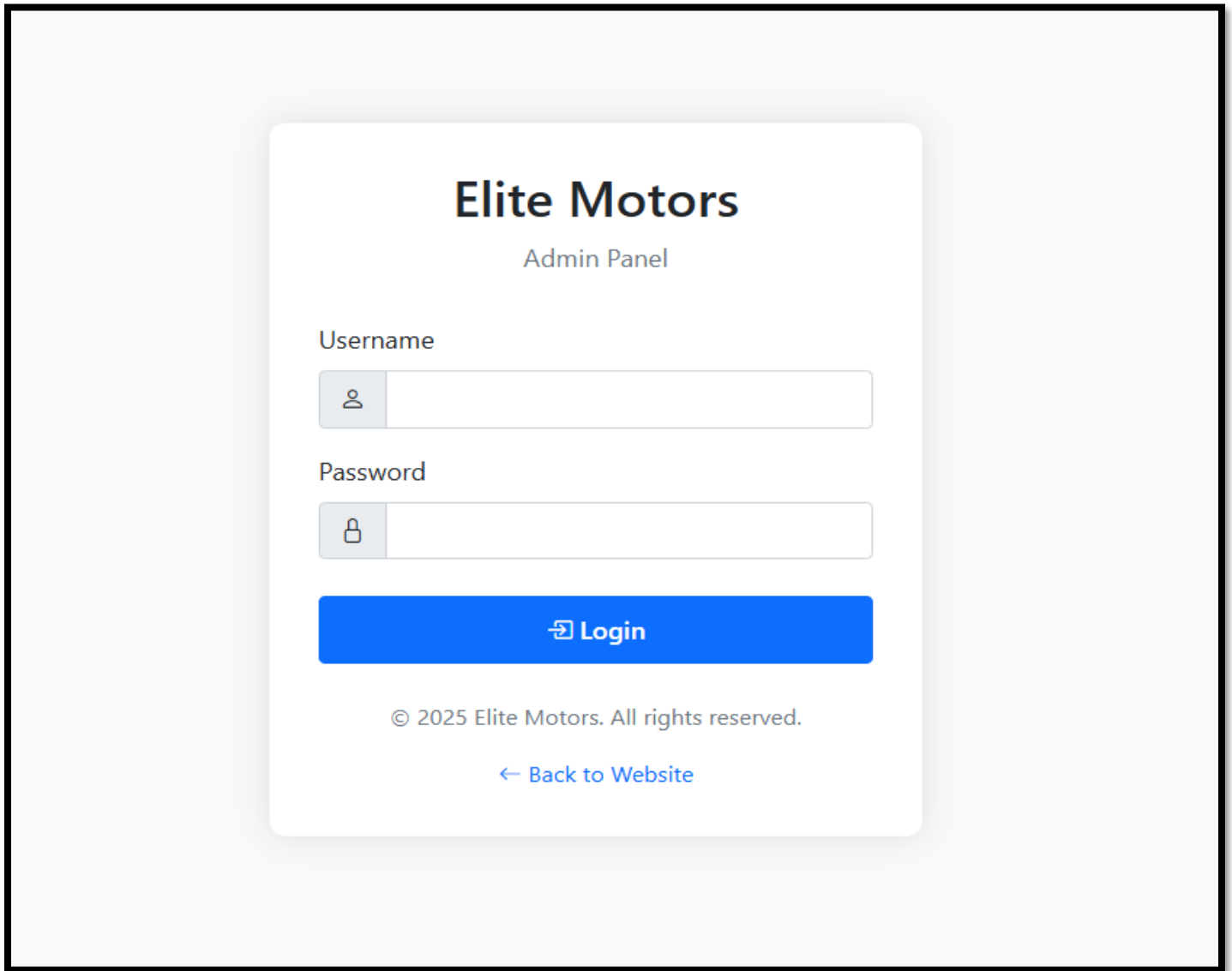
Car	Price	Status	Inquiries	Listed On	Actions
 2000 Maruti omni 70 miles	₹60,000	ACTIVE	0 Inquiries	Oct 1, 2025	Actions
 2010 Mercedes AMG 40 miles	₹10,000,000	ACTIVE	0 Inquiries	Oct 1, 2025	Actions
 2018 audi q7 60 miles	₹3,600,000	PENDING	0 Inquiries	Oct 1, 2025	Actions
 2020 Dodge Challenger 90 miles	₹35,000,000	ACTIVE	0 Inquiries	Oct 1, 2025	Actions
 2022 Lamborghini Huracan H2 50 miles	₹100,000,000	ACTIVE	0 Inquiries	Oct 1, 2025	Actions

Showing 1 to 5 of 5 entries

Previous 1 Next

❖ Admin Panel Screenshots

- Admin Login Page



The image shows a login page for 'Elite Motors'. The page has a light gray background with a white login card in the center. The card contains the title 'Elite Motors' and subtitle 'Admin Panel'. Below these are two input fields: 'Username' with a person icon and 'Password' with a lock icon. A blue 'Login' button is positioned below the password field. At the bottom of the card, there is a copyright notice '© 2025 Elite Motors. All rights reserved.' and a link '← Back to Website'.

Elite Motors
Admin Panel

Username

Password

[Login](#)

© 2025 Elite Motors. All rights reserved.

[← Back to Website](#)

Admin Dashboard

Elite Motors

Admin Panel

Dashboard

Users

Cars

Messages

Logout

Welcome, Dharmik Admin

Dashboard

TOTAL CARS

5

AVAILABLE CARS

5

TOTAL USERS

3

Recent Activity

Deleted user #3

Oct 1, 6:58 PM

Deleted car #8

Oct 1, 6:58 PM

Deleted user #3

Oct 1, 6:56 PM

Deleted car #8

Oct 1, 6:56 PM

View All Activities

Quick Actions

View Cars

View Users

Manage User page

Elite Motors

Admin Panel

Dashboard

Users

Cars

Messages

Logout

Welcome, Dharmik Admin

Manage Users

Users List

ID	USERNAME	NAME	EMAIL	PHONE	STATUS	JOINED	ACTION
6	yash	yash Prajapati	yash1@gamil.com	9714470286	Active	Oct 1, 2025	Delete
5	Nikhil	Nikhil Prajapati	Nikhil@gmail.com	9586728561	Active	Oct 1, 2025	Delete
1	Dharmik	Dharmik prajapati	dharmikghaskata9@gmail.com	9876543211	Active	Oct 1, 2025	Delete

■ Manage Car Page

The screenshot shows the 'Manage Cars' section of the Elite Motors Admin Panel. The left sidebar contains navigation links: Dashboard, Users, Cars, Messages, and Logout. The top right shows 'Welcome, Dharmik Admin' and a user profile icon. The main content area has a 'Manage Cars' header and three summary cards: 'TOTAL CARS' (5), 'APPROVED' (4), and 'PENDING APPROVAL' (1). Below these is the 'All Cars' section with the subtitle 'Manage all car listings in the system'. A yellow badge indicates 'Pending Approval 1'. Three car listings are shown: '2018 audi q7' (Price: ₹3,600,000, Mileage: 60 km), '2000 Maruti omni' (Price: ₹60,000, Mileage: 70 km), and '2010 Mercedes AMG' (Price: ₹10,000,000, Mileage: 40 km). Each listing has 'Edit' and 'Delete' buttons.

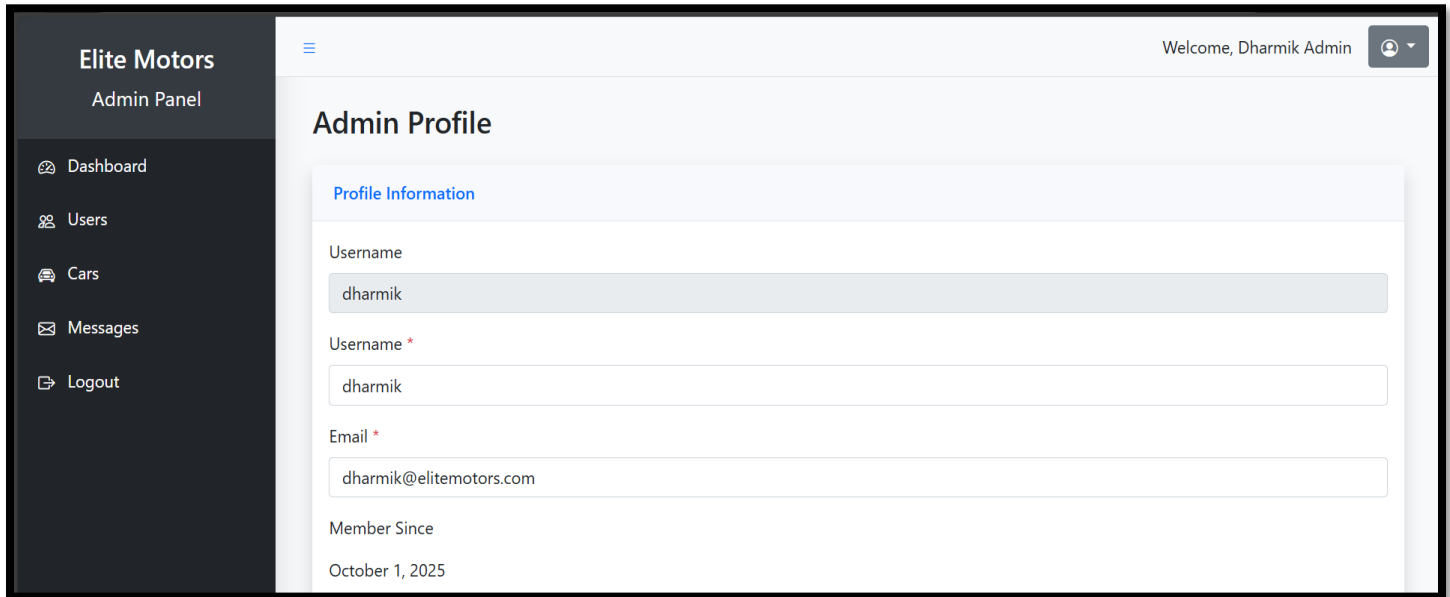
Car Model	Price	Mileage	Status
2018 audi q7	₹3,600,000	60 km	Approved
2000 Maruti omni	₹60,000	70 km	Approved
2010 Mercedes AMG	₹10,000,000	40 km	Pending Approval

■ User Message Contro page

The screenshot shows the 'Messages' section of the Elite Motors Admin Panel. The left sidebar is the same as the previous page. The top right shows 'Welcome, Dharmik Admin' and a user profile icon. The main content area has a 'Messages' header and a 'Contact Messages' section with a 'Total: 10' badge. Three messages are listed, all from 'Dharmik Ghaskata' (Email: dharmikghaskata9@gmail.com, Phone: 9586728561) under the 'About Car' category. The messages are: 'what is feature of dodge challenger car !!!!!' (Oct 1, 2025 10:03 PM), 'hello admin !!!!!' (Oct 1, 2025 5:36 PM), and 'hello admin !!!!!' (Oct 1, 2025 5:35 PM).

Message	Time
Dharmik Ghaskata: what is feature of dodge challenger car !!!!!	Oct 1, 2025 10:03 PM
Dharmik Ghaskata: hello admin !!!!!	Oct 1, 2025 5:36 PM
Dharmik Ghaskata: hello admin !!!!!	Oct 1, 2025 5:35 PM

Admin Profile page



Elite Motors
Admin Panel

Dashboard
Users
Cars
Messages
Logout

Welcome, Dharmik Admin

Admin Profile

Profile Information

Username
dharmik

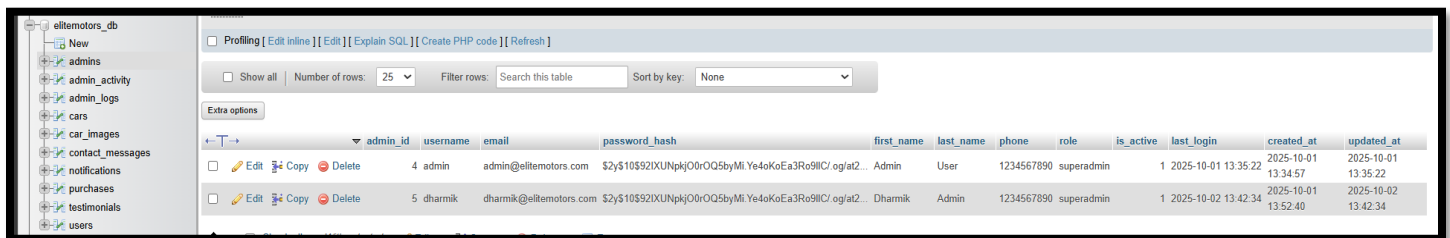
Username *
dharmik

Email *
dharmik@elitemotors.com

Member Since
October 1, 2025

Database

Admin database



admin_id	username	email	password_hash	first_name	last_name	phone	role	is_active	last_login	created_at	updated_at
4	admin	admin@elitemotors.com	\$2y\$10\$92IXUNpkj0OoRQ5byMi.Ye4oKoEa3Ro9llC/.og/at2...	Admin	User	1234567890	superadmin	1	2025-10-01 13:35:22	2025-10-01 13:34:57	2025-10-01 13:35:22
5	dharmik	dharmik@elitemotors.com	\$2y\$10\$92IXUNpkj0OoRQ5byMi.Ye4oKoEa3Ro9llC/.og/at2...	Dharmik	Admin	1234567890	superadmin	1	2025-10-02 13:42:34	2025-10-01 13:52:40	2025-10-02 13:42:34

Client/User database



user_id	username	email	password_hash	first_name	last_name	phone	address	profile_image	is_active	email_verified	verification_token	reset_token	reset_token_expire	last_login	created_at	updated_at
1	Dharmik	dharmikhasakata@gmail.com	\$2y\$10\$CaUd0mXAVH8qZKq35Rc.eRSDa88CUmzhBNm.EZNS...	Dharmik	prajapati	9876543211	NULL	NULL	1	0	NULL	NULL	NULL	2025-10-02 13:17:38	2025-10-01 13:19:19	2025-10-02 13:17:38
5	Nikhil	Nikhil@gmail.com	\$2y\$10\$K0H15vzRphIyEaxdRZuPluzsaQyenvHdc2SG55qU8...	Nikhil	Prajapati	9588728561	NULL	NULL	1	0	NULL	NULL	NULL	NULL	2025-10-01 18:20:31	2025-10-01 18:20:31
6	yash	yash1@gmail.com	\$2y\$10\$gETnsqz7JhQp275h10ETCuhEEmpRvY1tehuTyO9H9L...	yash	Prajapati	9714470286	NULL	NULL	1	0	NULL	NULL	NULL	2025-10-02 13:32:11	2025-10-01 19:22:25	2025-10-02 13:32:11

8. References

8.1 Core Technologies

Technology	Version	Purpose	Website
PHP	7.4+	Backend	php.net
MySQL	5.7+	Database	mysql.com
HTML5	-	Markup	MDN Web Docs
CSS3	-	Styling	MDN Web Docs
JavaScript	ES6+	Frontend Interactivity	MDN Web Docs

8.2 External Libraries

Library	Version	Purpose	Link
Bootstrap	5.2.0	CSS Framework	getbootstrap.com
jQuery	3.6.0	JavaScript Library	jquery.com
Font Awesome	6.0.0	Icons	fontawesome.com
DataTables	1.11.5	Table Plugin	datatables.net
PHPMailer	6.6.0	Email Sending	github.com/PHPMailer/PHPMailer

8.3 Development Resources

Documentation

- [PHP Manual](#)
- [MySQL Documentation](#)
- [Bootstrap Documentation](#)

Tutorials

- [W3Schools PHP Tutorial](#)
- [MDN Web Docs](#)

Tools

- [XAMPP](#) - Local development server
- [VS Code](#) - Code editor
- [Git](#) - Version control

Security References

- [OWASP PHP Security](#)
- [PHP Security Guide](#)
- [MySQL Security](#)

Performance

- [Google PageSpeed Insights](#)
- [WebPageTest](#)

The background is a warm, light beige color. It features several abstract, organic shapes in a slightly darker shade of beige. In the top-left and bottom-right corners, there are stylized leaf motifs in a golden-brown color. Small, thin, wavy lines are scattered across the background, adding a sense of movement and elegance.

Thank You

@EliteMotors